Frequency Sampling FIR Filter Design

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Key Concepts

1) Frequency sampling design minimizes the sum of the squared error between the desired and actual filter response at a discrete set of frequencies.

2) Frequency sampling design allows specification of arbitrary magnitude and phase response.

3) The filter frequency response at N frequencies is written as the product of an N-by-$M+1$ matrix and the vector of $M+1$ FIR filter coefficients. The matrix depends only on the frequencies and M.

4) The desired response can be satisfied exactly at up to $\frac{M+1}{2}$ distinct positive frequencies by solving a system of linear equations.

    a) The filter response is uncontrolled in between design frequencies.

    b) Erratic response may result if the desired response at the design frequencies is not smooth.

    c) Erratic response may result if the phase response at the design frequencies deviates significantly from linear phase corresponding to delay of $M/2$ samples.

5) A least-squares solution is used for more than $\frac{M+1}{2}$ design frequencies.

a) The solution no longer exactly satisfies the constraints at the design frequencies.

b) The criterion converges to the integral of the squared error as the N becomes large.

Video Screenshots

FIR filter

0:08

$$y[n] = \sum_{k=0}^{M} b_k x[n-k]$$

$$H(z) = \sum_{k=0}^{M} b_k z^{-k} \quad \Longrightarrow \quad H(e^{j\omega}) = \sum_{k=0}^{M} b_k e^{-jk\omega}$$

$$h[n] = \begin{cases} b_n & 0 \le n \le M \\ 0 & \text{otherwise} \end{cases}$$

Frequency sampling design: find b_k so $H(e^{j\omega_\ell}) \approx A_\ell e^{j\phi_\ell}$ for $1 \le \ell \le N$

Write $H(e^{j\omega})$ as an inner product of vectors

$$H(e^{j\omega}) = \sum_{k=0}^{M} b_k e^{-jk\omega} = \underbrace{\begin{bmatrix} 1 & e^{-j\omega} & e^{-j2\omega} & \cdots & e^{-jM\omega} \end{bmatrix}}_{\underline{d}(\omega)} \underbrace{\begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_M \end{bmatrix}}_{\underline{b}}$$

$$= \underline{d}(\omega)\,\underline{b}$$

Require conjugate symmetry for real $\underline{b}$

If $H(e^{j\omega_\ell}) = A_\ell e^{j\phi_\ell}$, then $H(e^{-j\omega_\ell}) = A_\ell e^{-j\phi_\ell}$

$$\begin{bmatrix} \underline{d}(\omega_\ell) \\ \underline{d}(-\omega_\ell) \end{bmatrix} \underline{b} = \begin{bmatrix} A_\ell e^{j\phi_\ell} \\ A_\ell e^{-j\phi_\ell} \end{bmatrix}$$

2 equations for each ℓ

$M+1$ unknowns in $\underline{b}$

Suppose $N/2$ constraint frequencies

$$\begin{bmatrix} \underline{d}(\omega_1) \\ \underline{d}(-\omega_1) \\ \underline{d}(\omega_2) \\ \underline{d}(-\omega_2) \\ \vdots \\ \underline{d}(\omega_{N/2}) \\ \underline{d}(-\omega_{N/2}) \end{bmatrix} \underline{b} = \begin{bmatrix} A_1 e^{j\phi_1} \\ A_1 e^{-j\phi_1} \\ A_2 e^{j\phi_2} \\ A_2 e^{-j\phi_2} \\ \vdots \\ A_{N/2} e^{j\phi_{N/2}} \\ A_{N/2} e^{-j\phi_{N/2}} \end{bmatrix}$$

full rank for distinct ω_ℓ

N equations

$M+1$ unknowns

$$\underline{D}\,\underline{b} = \underline{f}$$

$N \times M+1$

If $N \leq M+1$, we can exactly satisfy the design constraints

- Exact control of frequency response at up to $(M+1)/2$ frequencies

- No control of response at unspecified frequencies

  - Analogous to fitting polynomials $H(e^{j\omega}) = \sum_{k=0}^{M} b_k e^{-jk\omega}$

  - Specify gradual changes in $A_\ell e^{j\phi_\ell}$

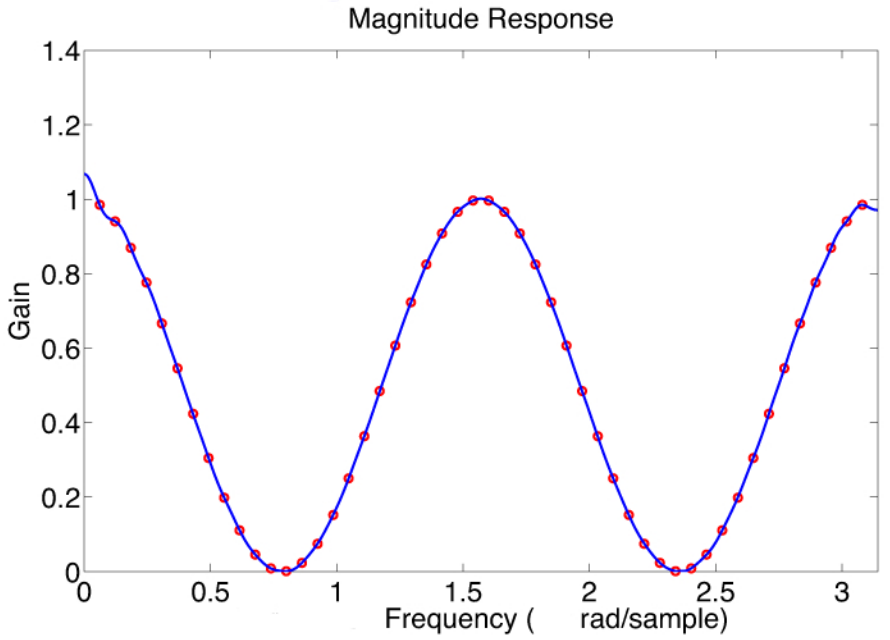
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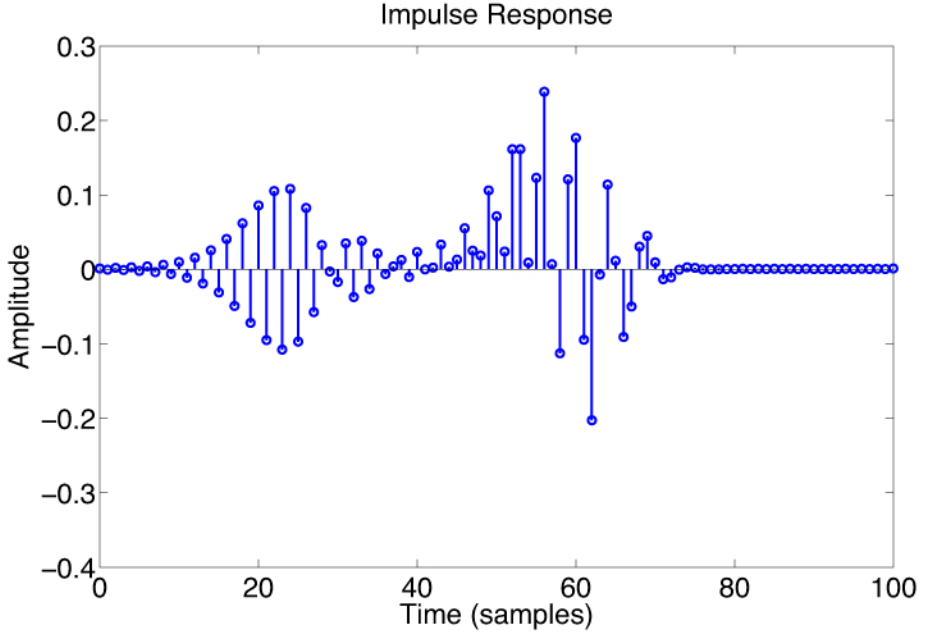
Example: $\omega_\ell = \frac{\ell\pi}{51}$, $\ell = 1, 2 \cdots, 50$

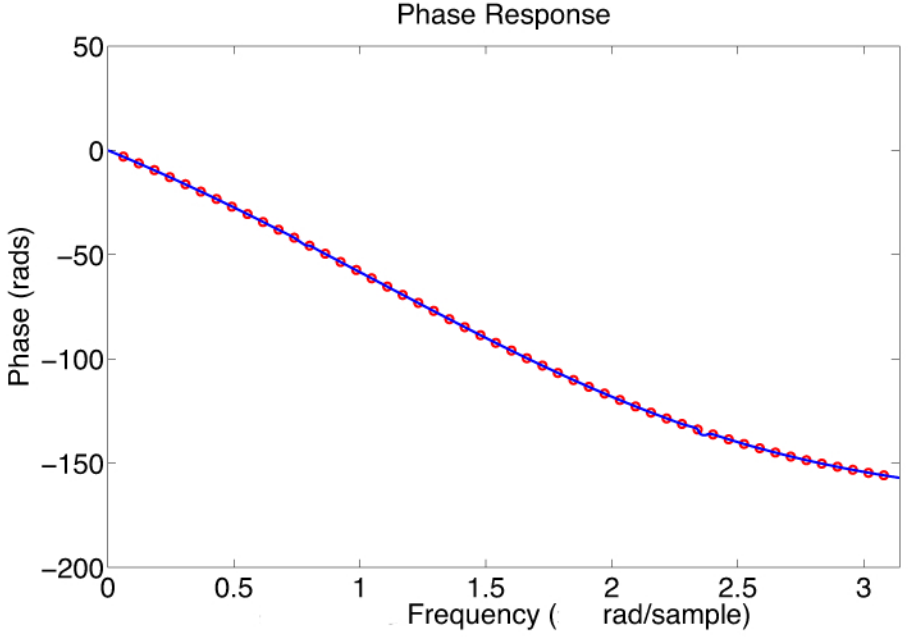
$A_\ell = (1 + \cos(4\omega_\ell))/2$

$\phi_\ell = -50\omega_\ell(1 + 0.2\sin(\omega_\ell))$

$M = 100$







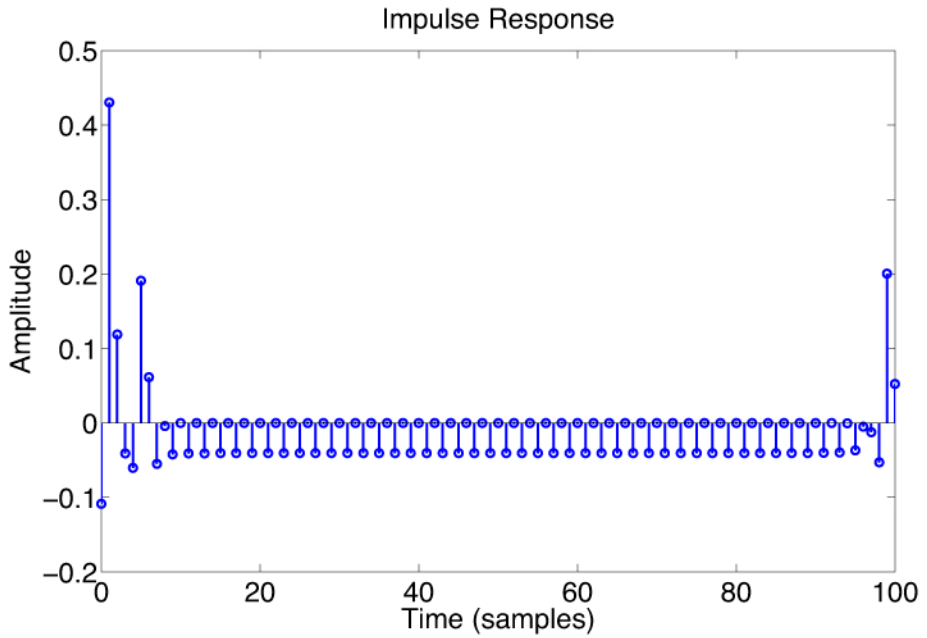
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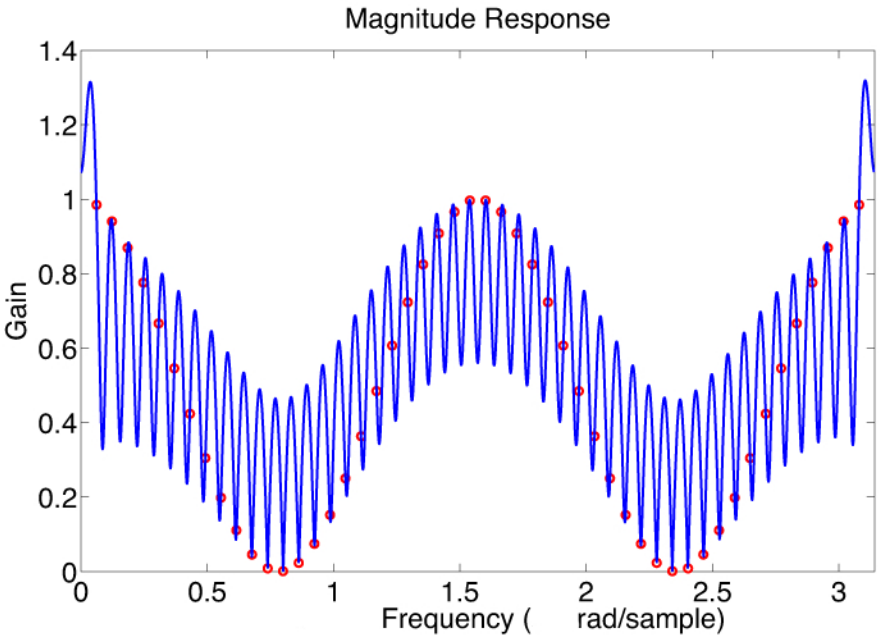
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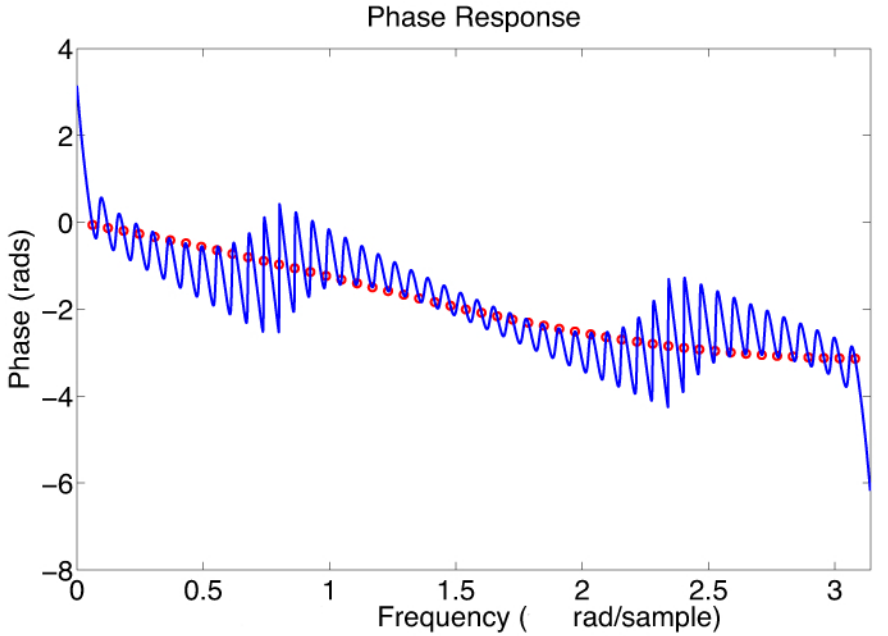
$A_\ell = (1 + \cos(4\omega_\ell))/2$

$\phi_\ell = -\omega_\ell(1 + 0.3\sin(\omega_\ell))$

$M = 100$







Least Squares Solutions: $N > M+1$

$\underline{D} \cdot \underline{b} = \underline{f}$

more equations than unknowns

$N \times (M+1) \cdot (M+1) \times 1 = N \times 1$

$$\min_{\underline{b}} |\underline{D}\underline{b} - \underline{f}|^2 \Rightarrow \underline{b} = (\underline{D}^H \underline{D})^{-1} \underline{D}^H \underline{f}$$

$(\cdot)^H$: complex conjugate transpose

- Don't obtain exact solution: $H(e^{j\omega_\ell}) \neq A_\ell e^{j\phi_\ell}$
- As $N \uparrow$, criterion approaches

$$\min_{\underline{b}} \int_{\omega=\pi} |H(e^{j\omega}) - A(\omega) e^{j\phi(\omega)}|^2 \, d\omega$$

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